Lab 03 – Network Segmentation (Paper or GNS3)

Industrial Security (Bachelor) • Maps to Section 03 • 45–60 min

LAB 03

Goal

Produce a Purdue diagram for a small water-treatment plant and a per-conduit rule table that an iDMZ firewall could enforce.

Step 1 – Draw

On a single sheet (or in `draw.io` / `excalidraw`) draw the architecture from Exercise 3.1 with every Purdue level labelled (L0–L5 and the iDMZ at L3.5).

Step 2 – Conduit table

For every conduit, write a row: *source, destination, protocol, port, direction.* Mark which conduit would benefit from a hardware data diode.

Step 3 – Optional: build it

In GNS3 or EVE-NG: three switches (Enterprise / iDMZ / Control), one pfSense, three Linux clients. From the Enterprise host, run `nmap -Pn 192.168.10.10`: it must return 0 open ports.

Deliverable

Hand in: the diagram and the conduit rule table. Screenshot of pfSense rules if you did the optional build.

Caution

The Docker stack does not enforce L3 isolation – this lab is intentionally design-focused.